



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
PRE-BOARD EXAMINATION

CLASS X
SUBJECT: PHYSICS (SET A)

FULL MARKS: 80
TIME: 2 HOURS

Maximum Marks: 80

Time allowed: Two hours

Answers to this Paper must be written on the paper provided separately.

You will not be allowed to write during the first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Section A is compulsory. Attempt any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets []

This paper consists of seven printed pages

SECTION A

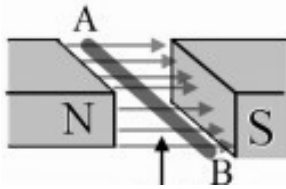
(Attempt all questions from this section)

Question 1

Choose the correct answers to the questions from the given options: [15]

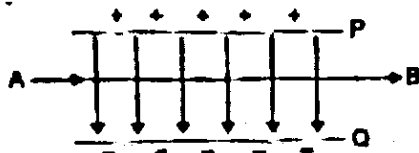
- (i) In an electrical cell while in use, the change in energy is from
a) electrical to magnetic
b) chemical to electrical
c) mechanical to electrical
d) electrical to mechanical
- (ii) Lenz's law is a consequence of the law of conservation of
a) charge
b) mass
c) energy
d) momentum
- (iii) Three equal resistances, when combined in series, are equivalent to $90\ \Omega$. Their equivalent resistance when combined in parallel will be
a) $270\ \Omega$
b) $30\ \Omega$
c) $810\ \Omega$
d) $10\ \Omega$
- (iv) Two blocks P and Q of different metals having their masses in the ratio 3 : 2 are given the same amount of heat. Their temperatures rise by the same amount. Ratio of their specific heat capacities is
a) 3 : 2
b) 9 : 4
c) 2 : 3
d) 4 : 9
- (v) A Ray of light suffers refraction through an equilateral prism. The deviation produced by the prism doesn't depend on the
a) angle of incidence
b) color of light
c) material of prism
d) area of the refracting surface

- (vi) The diagram below shows a free conductor AB is kept in a magnetic field and is carrying current from A to B. (To avoid confusion complete path of the circuit is not shown) The direction of the force experienced by the conductor will be



- a) up b) down c) towards N d) towards S
- (vii) A light ray enters from a denser medium to a rarer medium. With respect to the statement, choose the correct statement from the following
- The light ray bends towards the normal.
 - Angle of incidence is more than angle of refraction.
 - Frequency of the light remain same.
 - Speed of light remains unchanged.
- (viii) The critical angle for glass-air is 45° for the light of yellow color. The critical angle for green color is
- less than 45°
 - greater than 45°
 - equal to 45°
 - none of the above
- (ix) A Monochromatic light of wavelength λ gets refracted from vacuum to a medium of refractive index μ . The ratio of wavelength of incident and refracted wave is
- $1: \mu$
 - $\mu: 1$
 - $1: 1$
 - $\mu^2: 1$
- (x) A body is acted upon by two forces each of magnitude F , but in opposite directions. State the effect of forces if the forces act along same line of action.
- Resultant force is zero
 - Moment of force is zero
 - No motion will be produced
 - All the options are correct
- (xi) A ball is projected into the air with 100 J of kinetic energy which is transformed to gravitational potential energy at the top of its trajectory. When it returns to its original level after encountering air resistance, its kinetic energy is
- less than 100 J
 - more than 100 J
 - 100 J
 - insufficient information given
- (xii) A piece of glass immersed in a liquid, becomes invisible since
- both glass and liquid are colorless
 - both have identical refractive index
 - glass reflects the light transmitted by the liquid
 - both have same mass density
- (xiii) For a single movable pulley, if load is raised by a distance d , free end of the string is pulled up by
- the effort through a distance $2d$
 - the effort through a distance $4d$
 - the effort through a distance $d/2$
 - the effort through a distance d

- (xiv) Electric field is applied between two plates P and Q. A radioactive radiation AB passes through it as shown in the given figure. The ray is



- a) alpha ray b) beta ray c) gamma ray d) cathode ray.

- (xv) A sound wave is reflected from a brick wall. The reflected wave will have

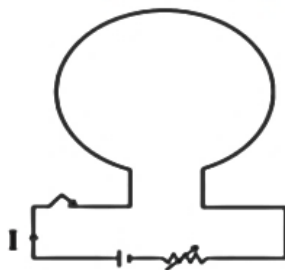
- a) same speed as that of the incident wave
b) same velocity as that of the incident wave
c) a greater amplitude compared to the incident wave
d) a shorter wavelength compared to the incident wave

Question 2

- (i) (a) How many strands are supporting the load in case of block and tackle system of 4 pulleys, where the effort is applied in a upward direction?
(b) Arrange the nuclear radiations in ascending order of biological damage caused by them.
(c) Which lens, thick or thin has greater focal length? [3]
- (ii) Calculate the power of an engine required to lift 1000 kg of coal per hour from a mine 360 m deep. [Take $g = 10 \text{ ms}^{-2}$] [2]
- (iii) The specific heat capacity of a substance A is $3,800 \text{ Jkg}^{-1} \text{ K}^{-1}$ and that of a substance B is $400 \text{ Jkg}^{-1} \text{ K}^{-1}$. Which of the two substances is a good conductor of heat? Give a reason for your answer. [2]
- (iv) The iron door of a building is 3 m broad. It can be opened by applying a force of 100 N normally at the middle of the door. Calculate the minimum force needed to open the door. [2]
- (v) Write any two conditions for hearing an echo. [2]
- (vi) An electric heater of power 100 W raises the temperature of 5 kg of a liquid from 25°C to 31°C in 2 minutes. Calculate the specific heat capacity of the liquid. [2]
- (vii) Why is it unsafe to connect a fuse to the neutral wire? [2]

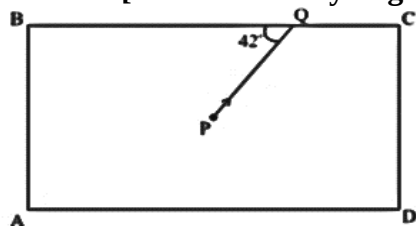
Question 3

- (i) The diagram below shows a loop of wire carrying current I:



- a) What is the magnetic polarity of the loop that faces us?
b) With respect to the diagram how can we increase the strength of the magnetic field produced by this loop? [2]

- (ii) How does an increase in the temperature affect the specific resistance of a
 (a) Metal and (b) Semiconductor? [2]
- (iii) State the conditions required for total internal reflection of light to take place. [2]
- (iv) A sound wave travelling in water has wavelength 0.4 m. Is this wave audible in air? Justify your answer. (The speed of sound in water = 1400 ms^{-1}) [2]
- (v) The diagram below shows a light source P embedded in a rectangular glass block ABCD of critical angle 42° . Complete the path of the ray PQ till it emerges out of the block. [Show necessary angles in the diagram.] [2]

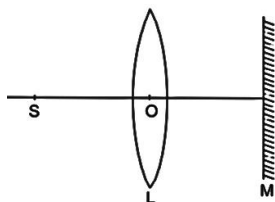


SECTION B

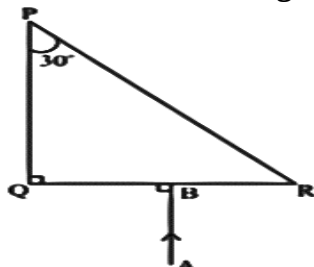
(Attempt any four questions)

Question 4

- (i) The following diagram shows a convex lens L, a plane mirror M and a point source of light S. Rays of light from the source S return to their point of origin.
 a) Complete the ray diagram to show this.
 b) What is the purpose of plane mirror M?
 c) What is the effect of moving the mirror M towards the lens L on the return of rays of light? [4]



- (ii) a) State two differences between mechanical and electromagnetic waves.
 b) Name the electromagnetic radiation used for satellite communication. [3]
- (iii) Complete the path of the ray AB through the glass prism in PQR till it emerges out of the prism. Given the critical angle of the glass as 42° . [3]



Question 5

- (i) An object is placed at a distance 24 cm in front of a convex lens of focal length 8 cm.
 a) What is the nature of the image so formed?
 b) Calculate the distance of the image from the lens.
 c) Calculate the magnification of the image. [4]

- (ii) A virtual and diminished image is formed when an object is placed between the optical centre and the principal focus of a lens.
- Name the type of lens which forms the above image.
 - Draw a ray diagram to show the formation of the image with the above stated characteristics. [3]
- (iii) a) A boy uses blue colour of light to find the refractive index of glass. He then repeats the experiment using red colour of light. Will the refractive index be the same or different in the two cases? Give a reason to support your answer.
- b) The refractive index of water is $\frac{4}{3}$ and that of glass is $\frac{3}{2}$. What is the refractive index of glass with respect to water? [3]

Question 6

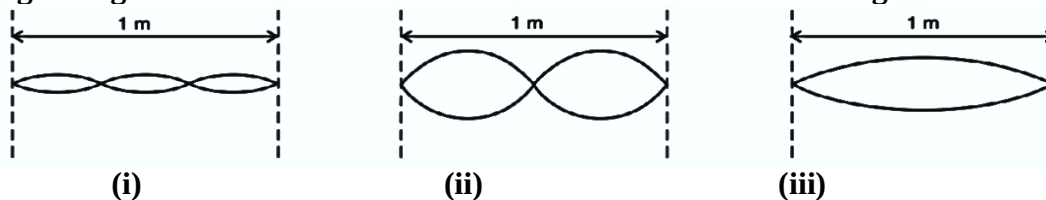
- (i) A simple pendulum having a bob of mass 500 g oscillating to and fro, rises to a maximum height of 5 cm above the mean position.
- Find the velocity of the bob at the mean position.
 - State the principle used to reach the conclusion.
 - Find the total energy of the bob. [Take $g = 10 \text{ ms}^{-2}$] [4]
- (ii) The following diagram shows a steering wheel of radius 0.25 m which is rotated clockwise by a torque 10 Nm produced due to a pair of equal and opposite force F.
- Calculate the magnitude of force F.
 - Show the force/forces in the diagram with their point of application [3]



- (iii) A block and tackle system of pulleys has velocity ratio 4.
- Draw a neat, labelled diagram of the system indicating clearly the points of application and direction of load and effort.
 - What will be its velocity ratio, if the weight of the movable block is doubled? [3]

Question 7

- (i) The diagrams given below show three modes of vibrations of a 1 m long stretched string.

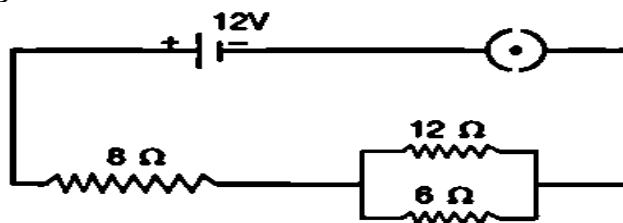


- If the frequency of the note produced in (i) is 'f', write down the expressions for frequency in cases (ii) and (iii) in terms of 'f'.
 - Find the wavelength of the vibration produced in (iii)
 - Which of the vibrations shown above will produce shrillest note? [4]
- (ii) A nucleus ${}_{11}^{24}\text{Na}$ is β -radioactive
- Write the equation representing β -decay.
 - What general name is given to the product with respect to ${}_{11}^{24}\text{Na}$? [3]

- (iii) A man standing in front of a vertical cliff fires a gun. He hears the echo after 3.5 second. On moving closer to the cliff by 84 m, he hears the echo after 3 seconds. Calculate the distance of the cliff from the initial position of the man. [3]

Question 8

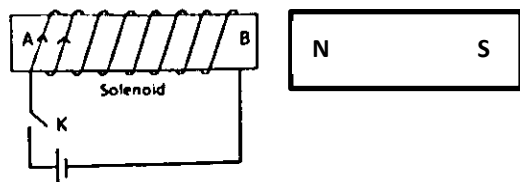
- (i) Three resistors $8\ \Omega$, $12\ \Omega$ and $6\ \Omega$ are connected to a 12 V battery as shown in the figure alongside. Calculate
- the current through the $8\ \Omega$ resistor.
 - the potential difference across the parallel combination of $6\ \Omega$ and $12\ \Omega$ resistors.
 - the current through the $6\ \Omega$ resistor. [4]



- (ii) a) What characteristics should a fuse wire have?
 b) Which wire in a power circuit is connected to the metallic body of an appliance? [3]
- (iii) A radioactive nucleus undergoes a series of decays, according to the following sequence
- $$X \xrightarrow{\beta} X_1 \xrightarrow{\alpha} X_2 \xrightarrow{\alpha} X_3$$
- If the mass number and atomic number of X_3 are 172 and 69 respectively, what are the mass number and atomic numbers of X ? [Rough working should be shown]
 - What will happen to the composition of nucleus X_3 , if it emits gamma radiation? [3]

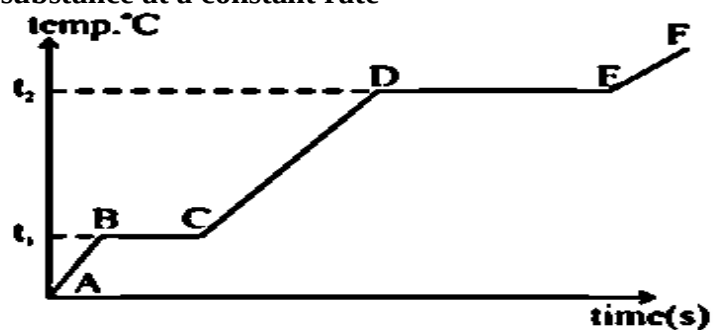
Question 9

- (i) The diagram in the figure shows a small magnet placed near a solenoid AB with its north pole N near the end B. Current is switched on in the solenoid by pressing the key K.
- What will be the observation wrt the movement of the magnet?
 - If the battery is replaced by a galvanometer and the magnet is moved away from the coil, what will be observed? Define the phenomenon observed.
 - Will there be any difference noticed in the galvanometer reading, if both coil and magnet are moved in the same direction at the same speed? [4]



- (ii) 200 g of water at 50.5°C is cooled down to 10°C by adding 'm' gram of ice cubes at 0°C in it. Find m. Take specific heat capacity of water = $4.2\text{ Jg}^{-1}\text{C}^{-1}$ and specific latent heat of ice = 336 Jg^{-1} [3]

(iii) The diagram below shows the change of phases of a substance on a temperature vs time graph on heating the substance at a constant rate



- a) What are the boiling and melting points of the substance?
- b) Why is the slope of AB greater than the slope of CD?
- c) Why are few segments of the graph parallel to the time axis?

[3]